Vipransh Ojha

OBJECTIVE

Committed to deliver accurate and actionable data by research and analyzing results. Demonstrating straightforward approach to problem-solving and executing tasks with efficiency and reliability.

SKILLS

- App Development (Flutter)
- Unity [Framework]
- Neural Networks
- Natural Language Processing (NLP)
- TensorFlow
- Pytorch
- Scikit-learn
- Python
- C++
- JAVA and SpringBoot
- Swift
- UI/UX Designing
- HTML and CSS
- Research paper writing
- Statistical analysis
- Analytical Thinking
- Teamwork and Collaboration

EXPERIENCE

FEBRUARY 2024 – APRIL 2024

Dr. Sumit Mittal & Dr. Satyam Ravi | School of Applied Science and Language | VIT Bhopal

Molecular Visualization Tool

- **Aim:** To develop a tool for visualizing molecular structures to enhance understanding in computational chemistry and related fields.
- Created the **Input Builder** module, enabling users to construct and modify molecular input data seamlessly.
- Focused on user-friendly interface design and efficient data parsing to support complex molecular datasets.
- Contributed to streamlining the visualization process, making it accessible to both researchers and students.
- Technologies Used: Python, PyMOL, OpenGL, Tkinter.

NOVEMBER 2023 – DECEMBER 2023

Dr. Pranjali Malviya | School of Computing Science and Engineering | VIT Bhopal

COVID-19 RNA Mutation Prediction Using Machine Learning

- **Aim:** To Process a large .fasta file containing RNA sequence data of the COVID-19 virus to analyze potential mutations.
- Trained a machine learning model to predict possible RNA sequence mutations, leveraging statistical patterns and biological insights.
- Implemented preprocessing techniques, including sequence encoding and feature extraction, to prepare data for model training.
- Applied classification/regression models and evaluated performance using metrics like accuracy, precision.
- Technologies Used: Python, Scikit-learn, Keras, TensorFlow, Pandas, Matplotlib, NumPy

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PROJECTS

1. Speech-to-Speech Translation System

Group Project

- Collaborated with a team to develop a real-time speech-to-speech translation system integrating speech recognition, neural translation, and text-to-speech synthesis for seamless multilingual communication.
- Contributed to the design and implementation of the following key features:
  - Real-time speech recognition using Openai Whisper and Google Speech-to-Text apis.
  - Neural translation leveraging advanced NLP models for high-accuracy language conversion.
  - Natural-sounding voice generation using Tacotron or similar TTS technologies.
  - Support for multiple languages with customizable voice attributes (e.g., pitch, speed, tone).
- Collaborated on modular pipeline design for low-latency and efficient real-time communication.
- Technologies Used: Python, Openai Whisper, Google Speech-to-Text API, Neural Machine Translation, Tacotron, Real-Time Frameworks.

2. SMS LINK VERIFIER APP

- Developed an android app that reads incoming SMS messages to identify and extract urls.
- Implemented a url verification system to determine whether links are safe or potentially malicious, leveraging real-time security checks.
- Integrated external apis and reliable databases to provide users with the source of verification for each link.
- Enhanced user safety by alerting them to potential phishing or scam links, contributing to proactive cybersecurity awareness.
- Technologies used: JAVA, SpringBoot, Gradle, Virustot

3. Hand Gesture-Controlled Mouse Interface

- Developed an innovative hand gesture recognition system using Python, OpenCV, and MediaPipe, enabling intuitive control of a computer mouse.
- Implemented a real-time video processing pipeline for hand tracking and landmark detection, utilizing MediaPipe's Hand Tracking API.
- Designed functionality to:
    - Control mouse movement through the index fingertip position mapped to screen dimensions.
    - Perform mouse clicks by detecting proximity between the thumb and index finger.
- Created a seamless user interface that operates efficiently at high frame rates, ensuring a responsive and engaging experience.
- Incorporated robust handling of dynamic hand gestures to differentiate between movement and clicking actions.
- Technologies Used: Python, OpenCV, MediaPipe, PyAutoGUI

4. FLAPPY BIRD AI SIMULATOR

- Designed and developed a flappy bird ai game using python, pygame, and neat (neuroevolution of augmenting topologies), combining game development with machine learning.
- Built a fully functional flappy bird game environment featuring bird animation, gravity mechanics, dynamic pipe generation, collision detection, and scoring system.
- Integrated a neat-based ai model to train virtual birds to play the game by evolving through generations, optimizing fitness scores for higher performance.
- Visualized live training performance and generation progress using matplotlib for real-time score plotting across generations.
- Implemented advanced features including:
    - Dynamic adjustment of game speed for performance analysis.
    - Real-time visual feedback on decision-making (e.g., jumping based on pipe positions).
    - Persistent high-score tracking and genetic optimization for improved ai performance.
- Technologies used: Python, Pygame, NEAT, Matplotlib, Multithreading

5. AUCTION PLATFORM

- Built a fully functional auction website where users can list items, place multiple bids, and sell products, fostering a dynamic and engaging marketplace experience.
- Developed core features such as user registration, secure login, item listing, bidding functionality, and public comment sections to enhance interactivity.
- Enhanced usability with an intuitive interface and streamlined navigation for easy item management and bidding processes.
- Technologies used: Python (Flask/Django), HTML, CSS (bootstrap), JavaScript, SQL, SQLite

EDUCATION

- VIT Bhopal University
  Bachelor’s Degree in Computer Science and Engineering (with specialization in AI&ML)
  Currently in 3rd Semester (2nd Year)

- Sardar Patel Vidyalaya, New Delhi
  Class X & XII (CBSE Board)